# Kurapati Srinivasa Abhiram

Indian Institute of Science Education and Research, Pune | Indian Institute of Technology Hyderabad

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## RESEARCH INTERESTS

Experimental Condensed Matter Physics, Functional Material Synthesis and Characterisation, Transport & Mechanical Property Measurements, Correlative Microscopy Techniques, Science Education.

## EDUCATION

**Indian Institute of Technology Hyderabad (IIT-HYD)** **Hyderabad, India**
MS Thesis, Department of Materials Science and Metallurgical Engineering *May* 2025 – *june 2026*

**Indian Institute of Science Education and Research Pune (IISER)** **Pune, India**
BS–MS Dual Degree, Physics *June* 2026
Distinction in Masters Thesis Project

## PUBLICATIONS

**Srinivasa Abhiram**, Suresh Perumal. Band Engineering of I-V-$VI_2$ chalcogenides for Thermoelectric Power Generation (In preparation).

## RESEARCH PROJECTS

**[1-A] Exploring III-VI Semiconductors for Functional Applications** *Aug 2026– present*
**Novel Materials and Thermeoelctrics laboratory, NCUE, Taiwan** ***Prof. Chia Ji Liu***
Extensively working on Bismuth – Tellurium based Semiconducting materials for Thermoelectric Device level application, Gaining expertise in influence of PEDOT:PSS influnece on Room Temparature synthesesis material.

**[2-A] Exploring III-VI Semiconductors for Functional Applications** *Dec 2025– June 2026*
**Laboratory for Energy and Advanced Devices (LEAD), IIT Hyderabad** ***Dr. Suresh Perumal***
Synthesised III-VI semiconducting materials and studies their structural, optical, and thermal characterization performed. Utilisation of these samples for functional applications are currently being explored.

**[2-B] Band Engineering of I-III-$VI_2$ for Thermoelectric Power Generation** *May 2025–Apr 2026*
Synthesised Cu-Sb-Se based chalcogenides via solid-state synthesis and densification through hot pressing. Te alloying was explored to tune thermoelectric properties; in-depth characterization was performed across structural, optical, chemical, and thermoelectric properties.

**[3] Encapsulation of Metal Oxides in CNTs** *Jan – April 2025*
**Exp. Condensed Matter Physics, Nanomagnets & Spintronics, IISERP** ***Dr. Ashna Bajpai***
Designing of CNT's via CVD (single and multi-walled) and encapsulated metal oxides; sample property analysis with AC susceptibility measurements and Raman spectroscopy.

**[4-A] Improvement of $zT$ in $p$-type Chalcogenide Material** *Apr –Dec 2024*
**Quantum Material Growth and Property Measurement Lab, IISERP** ***Prof. Surjeet Singh***
Synthesis and $zT$ measurements via Co-doping at the Sb and Te sites with Mn, Zn and Se respectively, in Ag-Sb-Se/Te-based Chalcogenide semiconducting materials through solid-state synthesis.

**[4-B] $n$-type Thermoelectric Half-Heusler Material** *Apr – Dec 2024*
Explored $n$-type TaFeSb Half-Heusler compounds via arc melting and powder synthesis; phase purity was identified as

a key challenge, with concurrent analysis of antisite disorder effects on $zT$ in $p$-type HH compounds.

**[5] Experimental Techniques in High Energy Physics** *Jan – Apr 2024*
**Experimental High Energy Physics Group, IISERP** ***Prof. Seema Sharma***
Hands-on experience in electronics through tabletop experiments with detectors, with emphasis on calibration and data acquisition methods. Studied high-energy physics applications in medical imaging.

**[6] Analysis of Bacterial Contamination in Establishments** *Aug 2022–Dec 2022*
**Bugs & Drugs Laboratory, IISERP** ***Dr. Nishad Matange***
Investigated microbial colonization patterns and strains, highlighting the importance of hygiene practices.

**Other Projects**

Development of an open-source LaTeX (TikZ) schematic repository for experimental physics [🐱]
The Community Project: Developing web applications to streamline research workflows, especially TE [🌐]

## CONFERENCES & WORKSHOPS

**SPARC Workshop on Advances in Correlative Microscopy and Spectroscopy**, IIT H — Feb 2026
**Symposium on Magnetism and Spintronics**, IIT Hyderabad — Jan 2026
**79th IIM ATM**, IIT Hyderabad 🏆 *Best Poster Award* — Dec 2025
**Rietveld Refinement of X-Ray Diffraction (RRD)**, CACR Delhi — Nov 2025
**International Online Conference on Crystals (IOCC 2024)** — Sep 2024
**Sokendai Asian Winter School** — Jan 2023
**Introduction to Quantum Computing**, QKRISHI — Nov 2022

## AFFILIATION & PROFESSIONAL EXPERIENCE

**Member, Indian Institute of Metals** — 2025–Present
**Member, APS (62168980)** — 2023–Present
**NASA's Future Innovations in Gamma Rays (FIG SAG)** — 2024–Present
**Science Trust Project** — 2024–Present
**Student Ambassador, APS (American Physical Society)** — 2023–2025

## VOLUNTEERING

**Thesis Mentoring – B.Tech Student** — Apr 2026
Co-guided a B.Tech thesis student on the effect of Cd doping in GaSb. Providing support in experimental synthesis and analysis, data interpretation, and result optimization.

**Teaching Assistant – Energy Materials and Devices (MS5120)** — Jan 2026
Designed and developed teaching resources across select course modules.

**Volunteer – Poster Committee** — Nov 2025
Contributed in the planning and organising of a large-scale poster event comprising ∼900 posters.

## EXPERIMENTAL AND TRANSFERABLE SKILLS

Proficient in a range of synthesis techniques including Solid-State reaction, Combustion, Hydrothermal, and Molten-salt synthesis, across the full sample preparation and characterization workflow.

Bulk & Nano synthesis | Raman Spectroscopy | UV-Vis-Spectroscopy | EDS/EDX

DSC & TGA | XPS | Electrochemical Workstation | ZEM-3/LSR | BET | LFA | Micro & Nano-Indentation | Glovebox Operation | Arc-Melting | LaTeX | CVD | Induction & Resistive Hot Press | Screen Printing | GitHub

## RELEVANT COURSES

### Physics

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| Classical Mechanics I, II | Electrodynamics | Electronics I, II |
| Statistical Mechanics | Condensed Matter Physics | Physics Lab I–V |
| Particle Physics | Energy Materials and Devices | Non-Linear Dynamics |
| Quantum Mechanics I–IV | Optics | Physics at Nano-Scales |
| Mathematical Methods | High Energy Physics | Numerical Analysis |

### Software

- Origin Pro
- CASA XPS
- Crystal Diffract
- VESTA
- X'Pert HighScore
- Thermo-Calc
- ImageJ
- Full Prof
- GABI (LCA)
- Blender
- MS Office
- KiCad

### Courses/ Certifications

Introductory Chinese pronunciation and daily conversation — Sep 2026 (enrolled)
Density Functional Theory, Coursera — Sep 2026
DSC & TGA E-Learning Courses, NETZSCH — Feb 2026
Introduction to Machine Learning For Engineering Research (ML4ER), UW Madison, USA — Jan 2025

## LANGUAGES

- English (C1)
- Hindi (C1)
- Italiano (principiante)
- Telugu (Native)

## HOBBIES

- Web Design
- UI-UX
- Cooking
- Photography